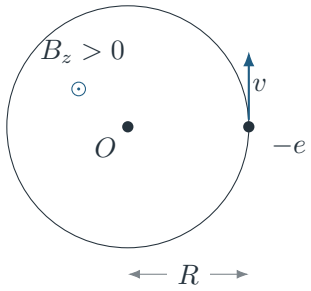
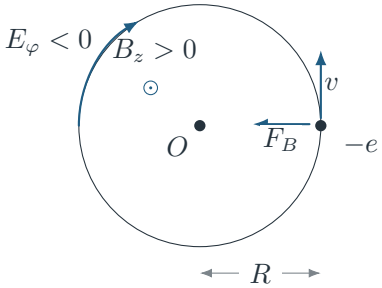
13-W1　有限动量注入的电子轨道



$$B_i \;\to\; B_f$$

图 13-9　题图　逆时针注入，局部场使电子向圆心偏转



$$B_i \;\to\; B_f$$

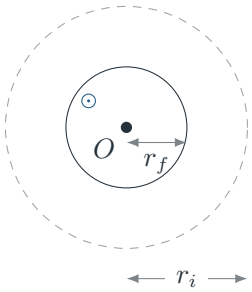
$$p \;=\; eRB_R$$

$$\Delta\overline{B} \;=\; 2\Delta B_R$$

图 13-10　解答分析图　轨道约束给出动量，电场由内部磁通的变化决定

13-W2　均匀增长磁场中的轨道收缩

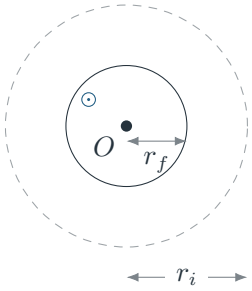
虚线：初轨道；实线：末轨道示意



$$B_i \;\to\; 4B_i$$

图 13-11　题图　均匀场缓慢增长，图中末轨道为初半径的一半

虚线：初轨道；实线：末轨道示意



$$B_i \;\to\; 4B_i$$

$$Br^2 \;=\; \text{const}$$

$$r_f \;=\; r_i/2, \quad p_f \;=\; 2p_i$$

图 13-12　解答分析图　瞬时圆轨道近似与正则角动量守恒共同确定收缩

13-W3　螺线管的电感与场能

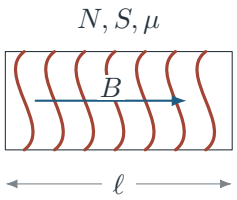
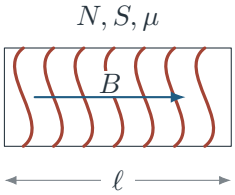


图 13-13　题图　长螺线管的匝数、截面和长度共同决定电感



$$\Psi = NBS$$

$$W_m = B^2 S\ell/(2\mu)$$

图 13-14　解答分析图　磁通链法与管内场能积分一致

13-W4　实心同轴内导体的电感

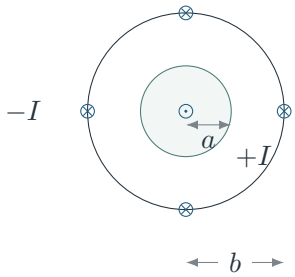
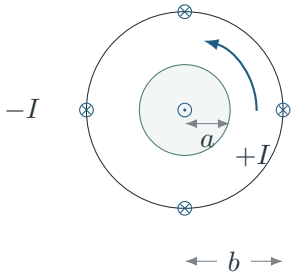


图 13-15　题图　实心内导体与薄外筒通有相反总电流



$$L' = L'_{\text{int}} + L'_{\text{ext}}$$

$r > b$ 时 $B = 0$

图 13-16　解答分析图　内导体与空隙分别储能，返回电流消除外部场

13-W5　接通 RL 回路后的能量分配

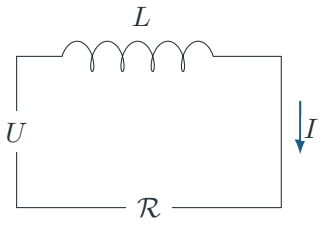
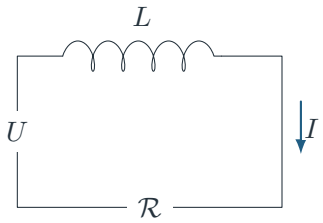


图 13-17　题图　恒压源从零电流建立 RL 回路



$$I = U(1 - e^{-t/\tau})/\mathcal{R}$$

$$W_s = Q + W_m$$

图 13-18　解答分析图　对实际电流积分电源功与焦耳热，差额为磁能

13-W6　原电流匀速增加时的次级电流

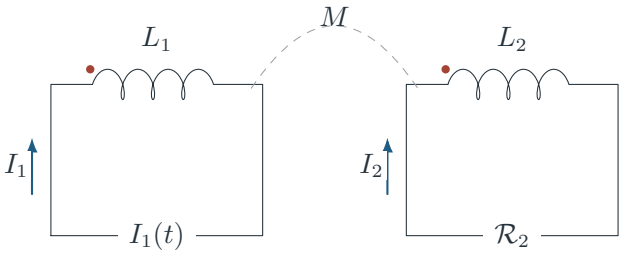
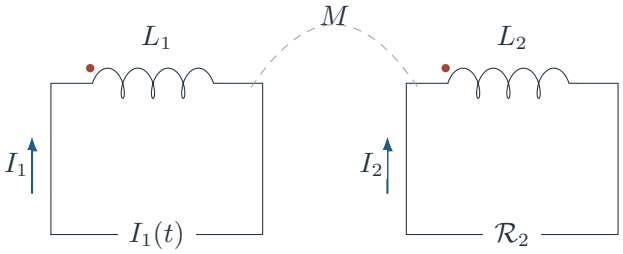


图 13-19　题图　点端规定两电流正向，原电流由外部装置给定



$$\dot{I}_1 = \alpha \Rightarrow I_2 < 0$$
$$\mathcal{R}_2 q_2 = -M\Delta I_1$$

图 13-20　解答分析图　次级电流反向，斜坡停止后仍继续衰减

13-W7　截面不同的共轴长绕组

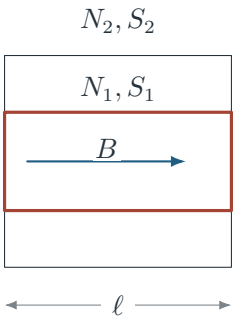
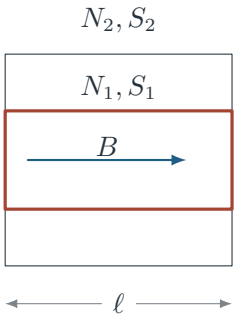


图 13-21　题图　红色内绕组截面较小，两长绕组共轴



$$M = \mu N_1 N_2 S_1 / \ell$$

$$k = \sqrt{S_1/S_2} < 1$$

图 13-22　解答分析图　互感使用共同截面，外绕组仍有未链接内绕组的磁通

13-W8　两个耦合回路的自由衰减

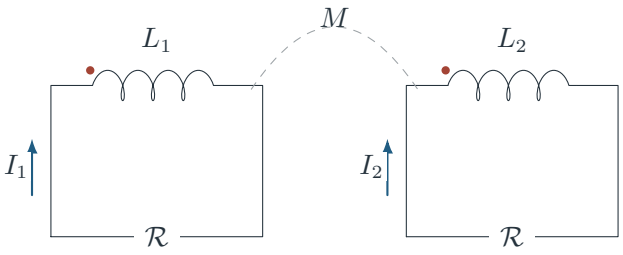
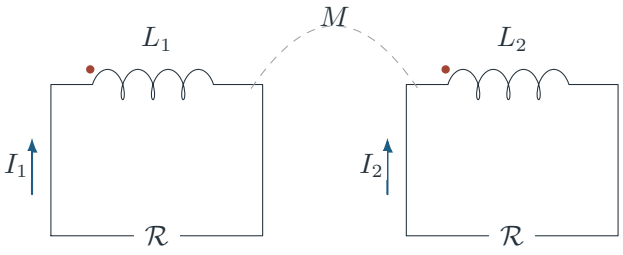


图 13-23　题图　两个闭合回路初始仅第一回路有电流



$$I_{\pm} = I_1 \pm I_2$$

$$\tau_{\pm} = (L \pm M)/\mathcal{R}$$

图 13-24　解答分析图　同向与反向电流组合对应不同衰减时间

13-W9　有互感线圈的等效电感

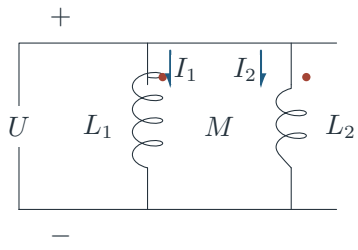
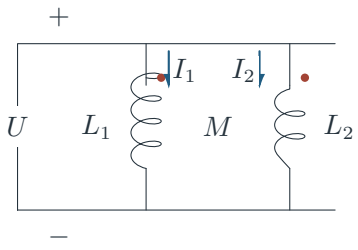


图 13-25　题图　同名端并联的两绕组端电压相同



$$L_+ = 19\,\mathrm{mH}$$
$$L_- = 7\,\mathrm{mH}$$
$$L_p = 27/7\,\mathrm{mH}$$

图 13-26　解答分析图　并联需联立磁通链方程，串联则按连接方向相加

13-W10　可变电感的拉力与供能

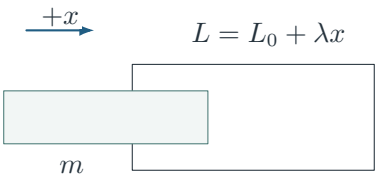


图 13-27　题图　部件向右进入，题设电感随位移增加

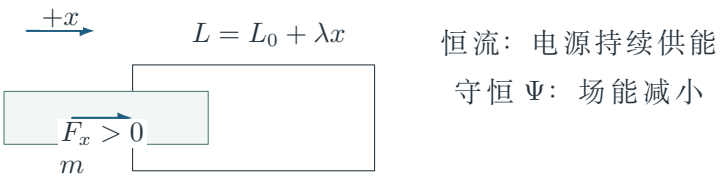


图 13-28　解答分析图　两种约束下均向电感增大方向受力，能量分配不同

13-W11　远距离共轴小线圈的吸引力

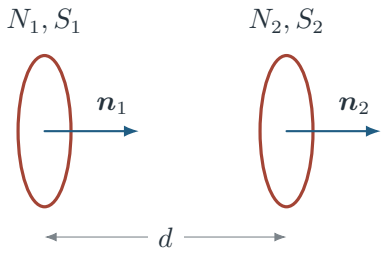


图 13-29　题图　共轴小线圈磁矩同向，间距远大于半径

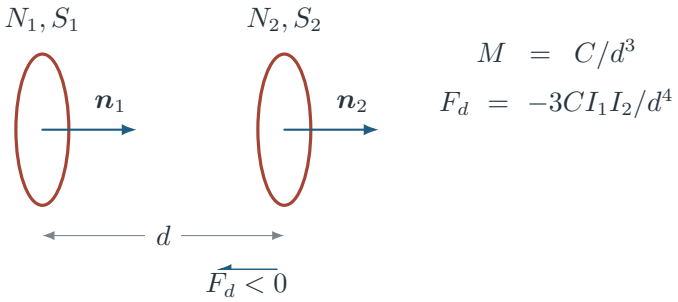
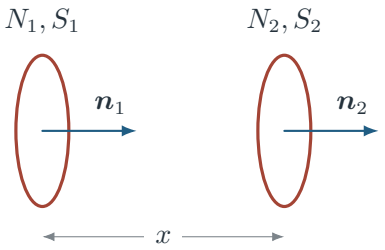


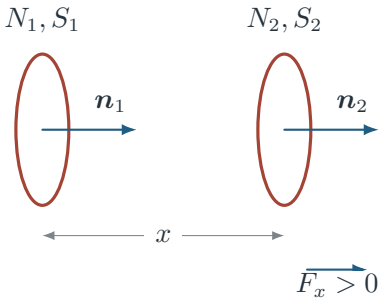
图 13-30　解答分析图　同向电流相吸，恒流源供给增加的场能和机械功

13-W12　闭合无电阻次级的排斥



$$I_1 = I_0, \quad \Psi_2 = 0$$

图 13-31　题图　次级无电阻闭合，法向箭头定义电流正向



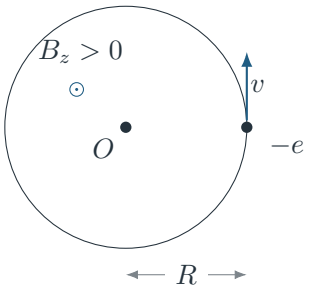
$$I_1 = I_0, \quad \Psi_2 = 0$$

$$I_2 = -MI_0/L_2$$

$$L_{\text{eff}} = L_1 - M^2/L_2$$

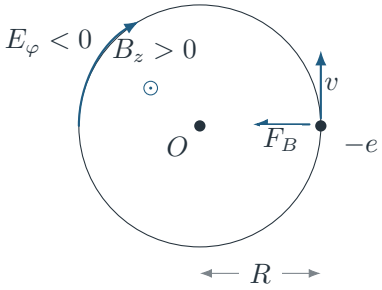
图 13-32　解答分析图　实际次级电流为负，两线圈相斥，原端口仍有漏感

13-T1　有偏置平均场的电子注入



$$\overline{B}_i \neq 2B_i$$

图 13-33　题图　平均场与轨道场的初始比例未预先指定



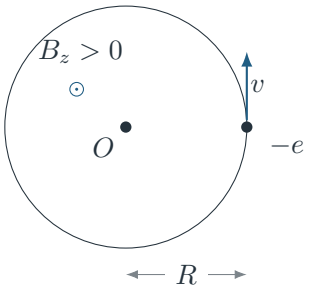
$$\overline{B}_i \neq 2B_i$$

$$\overline{B} - 2B_R = \text{const}$$

$$P_\varphi = eR^2(B_i - \overline{B}_i/2)$$

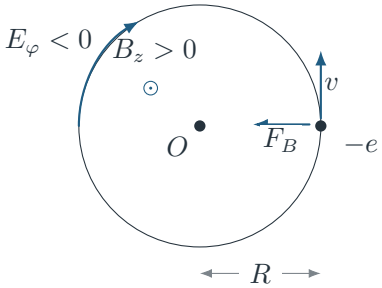
图 13-51　解答分析图　守恒常数由注入状态确定，平均场偏置可以保留

13-T2　每圈能量增量与瞬时功率



$$\dot{B}_R = \beta$$

图 13-34　题图　半径固定，轨道场以恒定斜率增加



$$\dot{B}_R = \beta$$

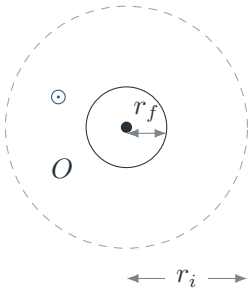
$$\Delta K_{\text{turn}} = 2\pi e R^2 \beta$$

$$P = eR\beta v$$

图 13-52　解答分析图　每圈增量与圈频相乘，恢复电力瞬时功率

13-T3　均匀场压缩时的动能

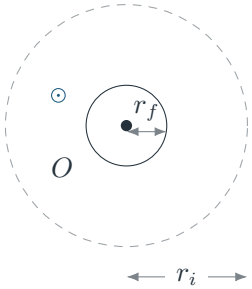
虚线：初轨道；实线：末轨道示意



$$B_f = 9B_i$$

图 13-35　题图　同心轨道的末半径为初半径的三分之一

虚线：初轨道；实线：末轨道示意



$$B_f = 9B_i$$

$$r_f = r_i/3$$

$$p_f = 3p_i$$

图 13-53　解答分析图　先求半径和动量，再按相对论关系求动能

13-T4　矩形截面环形线圈

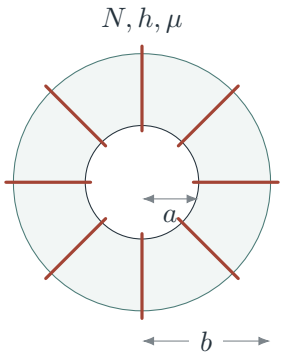
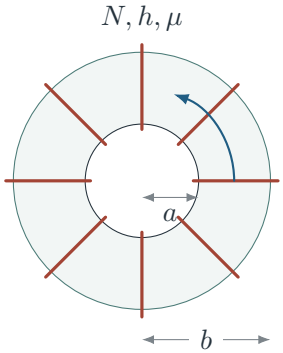


图 13-36　题图　环形线圈的矩形截面高度为 h，图为俯视



$$B(r) = \mu N I/(2\pi r)$$

$$\mathrm{d}V = 2\pi r h\,\mathrm{d}r$$

图 13-54　解答分析图　磁通与场能均保留径向变化的磁场

13-T5　径向分布电流的内部电感

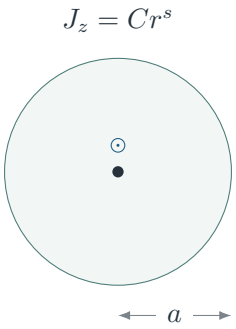
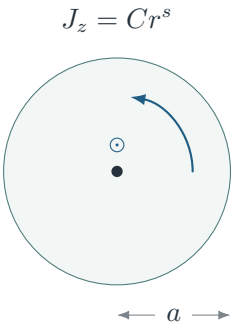


图 13-37　题图　电流沿纸外方向，径向密度由题设幂函数给定



$$I_{\text{enc}} = I(r/a)^{s+2}$$

$$L'_{\text{int}} = \mu/[4\pi(s+2)]$$

图 13-55　解答分析图　先积分包围电流，再用内部场能求电感

13-T6　有厚度的同轴返回导体

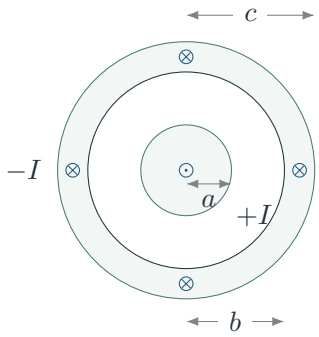
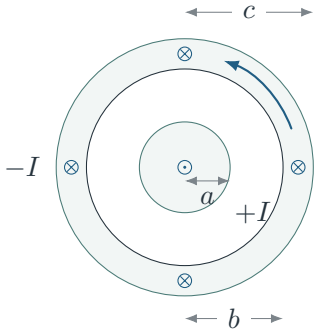


图 13-38　题图　实心内导体半径 a，外返回导体占据 b 到 c 的有限厚度



$$I_{\text{enc}} = I(c^2 - r^2)/(c^2 - b^2)$$

外壳场在 $r = c$ 降到零

图 13-56　解答分析图　外导体中的净包围电流逐渐减少，仍须计入内部场能

13-T7　趋肤深度与交流电阻

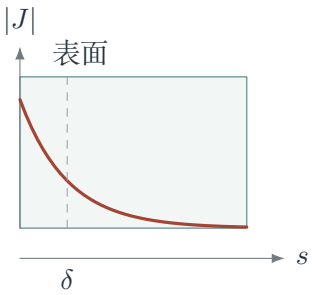
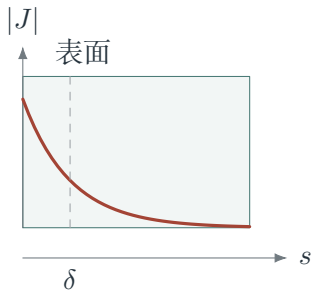


图 13-39　题图　深度 s 由表面向内量，曲线表示强趋肤幅值衰减



$$|J| \propto e^{-s/\delta}$$

$$\omega L'_{\mathrm{int}} \simeq \mathcal{R}'_{\mathrm{ac}}$$

图 13-57　解答分析图　表层电流的相位与幅值同时变化，内部阻抗含电阻和电感

13-T8　钳位放电的停止时间

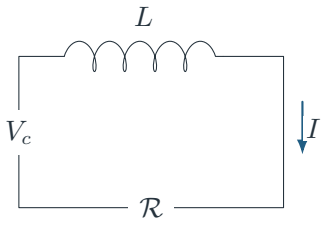


图 13-40　题图　钳位器件在放电电流阶段吸收恒定端电压的功率

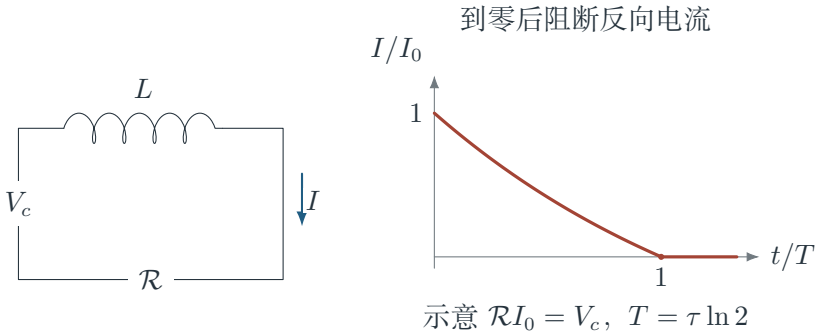


图 13-58　解答分析图　指数解到首次零电流为止，随后由器件阻断

13-T9　原电流往返变化的传输电荷

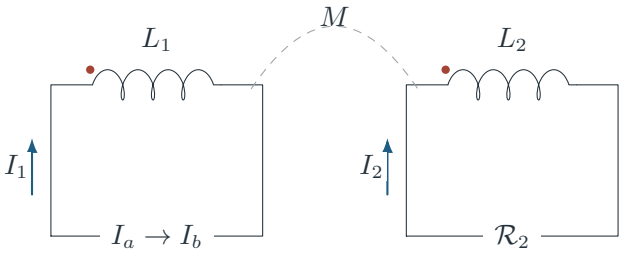
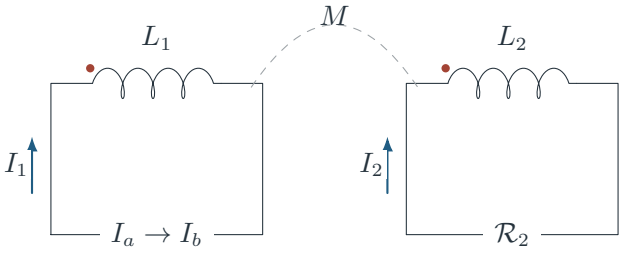


图 13-41　题图　原电流变化驱动闭合次级，初末次级电流为零



$$q_2 = -M(I_b - I_a)/\mathcal{R}_2$$

$$q_2 = 0 \text{ 不推出 } Q_2 = 0$$

图 13-59　解答分析图　代数电荷可抵消，电流平方的热量仍为正

13-T10　由三种接法测定自感和互感

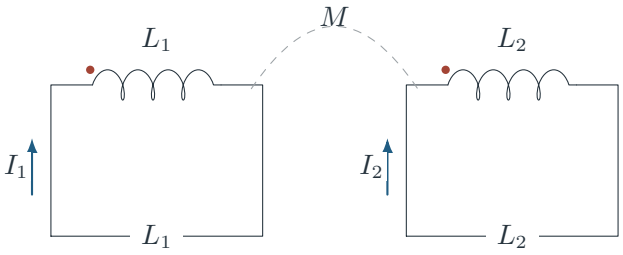


图 13-42　题图　两个绕组的同名端已标出，连接由题目指定

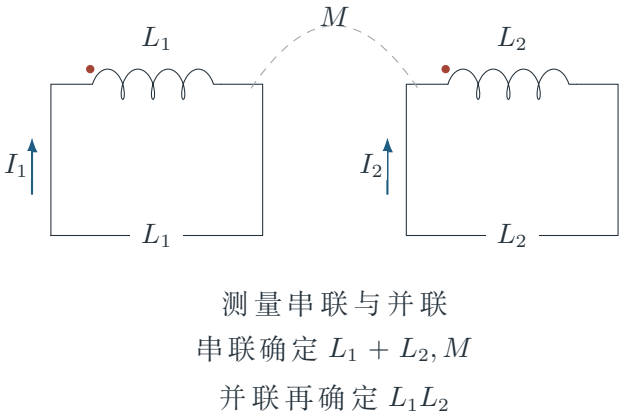


图 13-60　解答分析图　三种测量分别给出和、互感与乘积

13-T11　并联支路中的反向电流

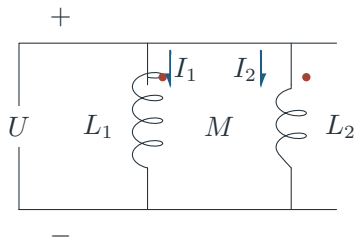
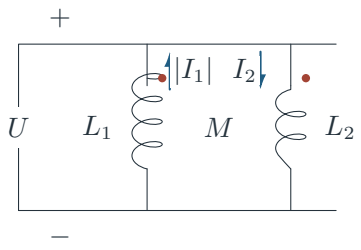


图 13-43　题图　箭头定义两支路的正向，初始电流都为零



$$I_1 < 0, \quad I_2 > 0$$
$$W_m = 600U^2t^2 > 0$$

图 13-61　解答分析图　实际第一支路电流向上，点端坐标中为负，矩阵正定且总磁能为正

13-T12　电阻不同的耦合回路

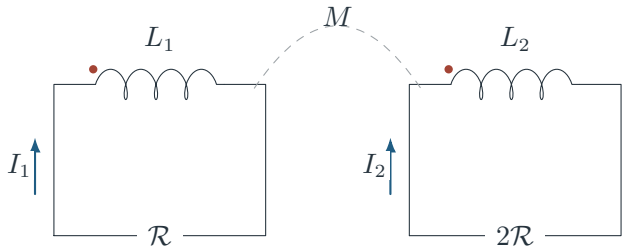
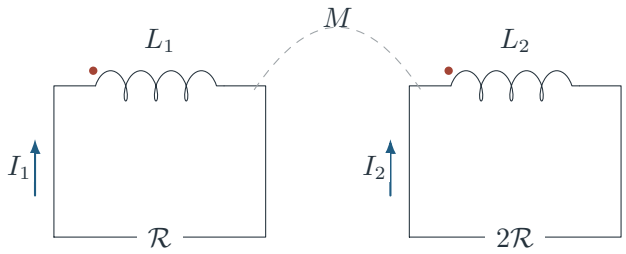


图 13-44　题图　两回路自感相同，电阻不同，初始仅原回路有电流



$$\lambda_s, \lambda_f \;>\; 0$$

$$q_1 \;=\; LI_0/\mathcal{R}$$

$$q_2 \;=\; MI_0/(2\mathcal{R})$$

图 13-62　解答分析图　衰减模态由特征方程求解，电荷由积分回路方程得到

13-T13　给定原电流时的最低磁能

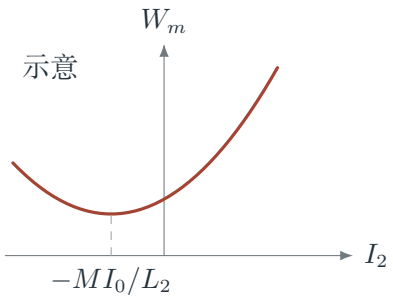
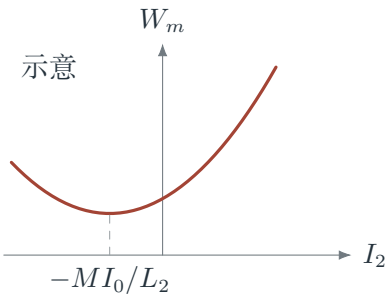


图 13-45　题图　原电流固定时，磁能关于次级电流是开口向上的抛物线



$\Psi_2 = 0$ 时达到最低能量

$$W_{\min} = L_{\text{eff}} I_0^2/2$$

图 13-63　解答分析图　顶点对应零次级磁通链，曲线高度表示剩余漏磁能

13-T14　转动线圈的平衡与小振动

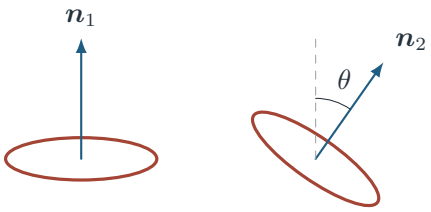


图 13-46　题图　两线圈法向夹角为 θ，互感随余弦改变

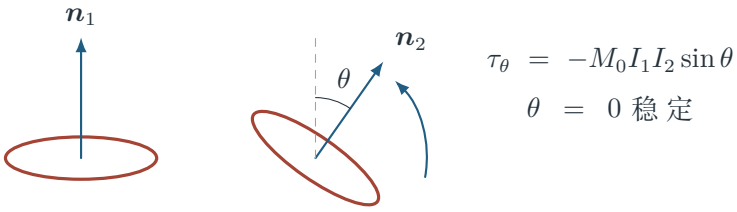


图 13-64　解答分析图　磁转矩使法向趋于同向，电源参与运动供能

13-T15　守恒磁通链推动自由部件

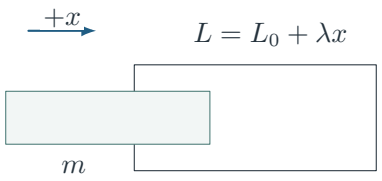
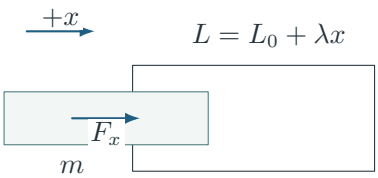


图 13-47　题图　无电阻回路使磁通链守恒，可动部件向右进入



$$\Psi = L_0 I_0$$

$$mv^2/2 + \Psi^2/(2L) = \text{const}$$

图 13-65　解答分析图　场能下降转为动能，电流随电感增加而减少

13-T16　三个等价线圈的互感限制

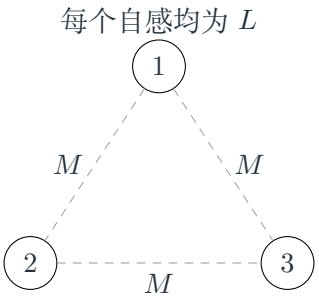
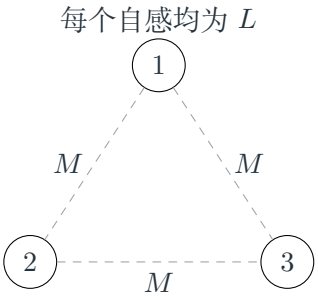


图 13-48　题图　三个等价线圈的每对互感都为 M，连线只表示耦合



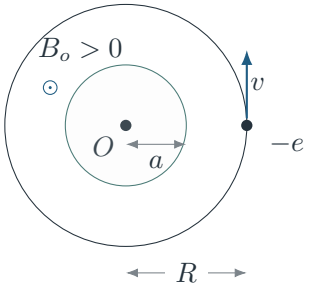
共同模态：$L + 2M$

差模态：$L - M$（二重）

$-L/2 \;\le\; M \;\le\; L$

图 13-66　解答分析图　整体能量还约束共同模态，两两上界不能代替它

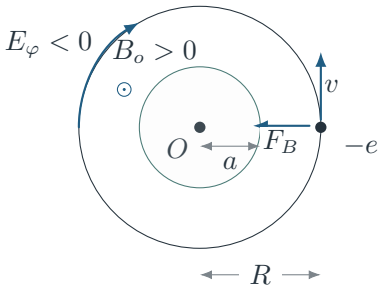
13-T17　分区磁场中的注入与加速（综合题）



内 区：B_c

环 区：B_o

图 13-49　题图　中心区半径 a，电子参考轨道半径 R，两区场增长率不同



内 区：B_c

环 区：B_o

$$\eta\kappa + (1-\eta)\beta = 2\beta$$

$$\mathcal{N} = \Delta K/(2\pi e R^2 \beta)$$

图 13-67　解答分析图　按面积加权磁通，参考轨道、动量与绕行圈数相互一致

13-T18　可动次级的制动与电源交换能量（综合题）

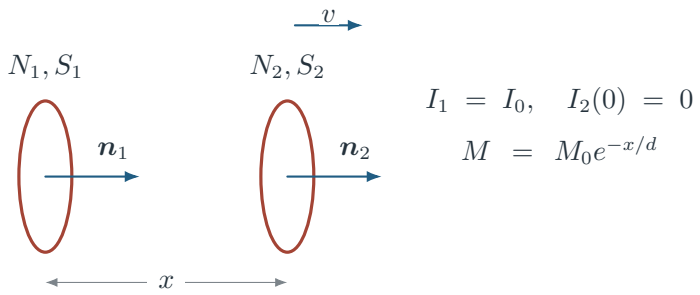


图 13-50　题图　外装置使间距增加，次级初始无电流但有原线圈磁通

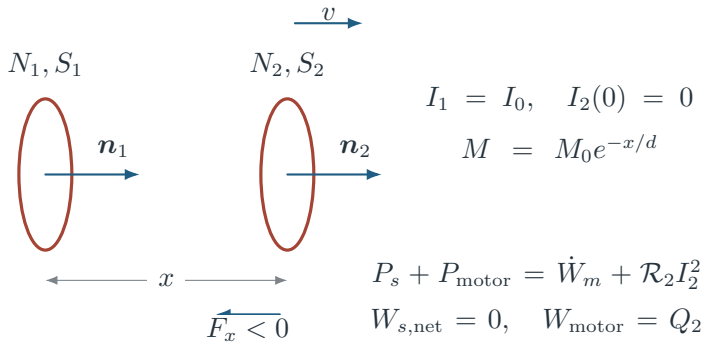


图 13-68　解答分析图　次级电流支持减少的磁通，磁力制动，电源先供能后回收